Markets for Experience Goods With Performance Uncertainty

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Abstract

This article analyzes the market for experience goods when the product's quality is uncertain. Unlike the case of certain product quality, in this setting, a customer who receives a bad product cannot automatically infer that the producer has "cheated." In this two-period model, producers select either high- or low-quality technologies, and consumers individually decide whether or not to rebuy the product in the second period. The sequential equilibria involve mixed strategies in producer "honesty" and buyer "trust," and the form of these strategies under conditions of monopoly and competition is investigated. Unreasonable equilibria are eliminated by applying a form of Kohlberg and Mertens' invariance criterion. Several extensions are analyzed, including the use of first-period discounts and an infinite horizon model obtained by independent repetition of the two-period case.

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Introduction

If product quality is costly and cannot be determined prior to consumption, why should consumers believe a seller's quality claims? Using Telser's (1980) idea of self-enforcing agreements, Klein and Leffler (1981) developed the now-standard solution to this problem: the price remains above the cost, providing high-quality sellers with rents, and consumers boycott any seller discovered cheating on quality. However, production technologies may have performance uncertainty; that is, the producer might not have perfect control over the quality of each unit. Consequently, the receipt of an inferior product might not always be a clear signal that a supplier is cheating. Given the existence of performance uncertainty, buyers receiving low-quality products do not know whether to blame their problems on malfeasance or on bad luck. This ambiguity can itself increase the seller's propensity to cheat.

In this article, we construct and analyze a model of an experience good market with performance uncertainty. The model features fully rational and homogeneous buyers and sellers-there are no arbitrary beliefs or "types" of player. The setup constrains a buyer's information to his personal experience, rules out any threats to boycott that are not credible

for each buyer, and precludes market equilibria in which a signal perfectly reveals the seller's quality.

The game admits a variety of "unreasonable" sequential equilibria, supported by what Grossman and Perry (1986) called incredible beliefs. Interestingly, there is a correspondence between the sequential equilibria that are supported by incredible beliefs in one form of the game tree and the Nash equilibria that involve incredible threats in another, trivially transformed, version of the game tree. Thus, we are able to rule out all but a single (reasonable) monopoly or competitive equilibrium by applying a specialization of Kohlberg and Mertens' (1986) criterion that equilibria survive trivial changes in the game tree.

The equilibria involve mixed strategies in seller "honesty" (high- versus low-quality) and buyer "trust" (rebuy versus boycott). Consequently, the model permits a richer exploration of comparative statics than can be accomplished with pure-strategy equilibria. We are able to examine how honesty and trust, as well as the price and social surplus, vary with changes in the parameters and between monopoly and competitive regimes. In addition, we investigate the impact of introductory price discounts, and we analyze an infinite horizon game generated by independent repetition of the two-period game.

Experience goods. An experience good is a product whose quality cannot be fully determined before it is purchased (Nelson, 1970), where quality refers to any valued attribute such as taste, efficiency, or durability. Most products are experience goods according to this description. For example, computer software, motion pictures, and most foods cannot be fully evaluated without "consuming" them. Durable products, such as automobiles, are experience goods with respect to their postpurchase reliability and durability. The services of teachers, lawyers, and doctors are also experience goods.

Given that costs increase with quality, incomplete prepurchase information can lead to opportunism, since profit-maximizing firms will always prefer to produce lower-quality goods for a given price and quantity. If consumers cannot determine the quality of a good until after they purchase it, the firm can cheat by claiming to supply high-quality goods, but delivering low-quality goods. Suspecting such behavior, consumers will refuse to buy; the resultant market failure causes society to incur welfare losses.

One mechanism that controls opportunism in experience good markets is reputation, i.e., consumers' beliefs about a firm's future behavior based on its past and present actions. [1] Telser (1980) and Klein and Leffler (1981) first showed that consumers can rationally believe a firm will supply high-quality experience goods if they are willing to pay prices above marginal costs and to make repeat purchases only so long as high-quality goods are supplied. The quality-assuring, or incentive, price must be large enough to make the value of repeated sales exceed the one-time gain from cheating, so that the firms will never find it optimal to cheat buyers.

A number of investigators have modified or extended this basic analysis. Allen (1984) analyzed a situation in which firms select higher levels of fixed costs in order to signal higher product quality. Shapiro (1982, 1983a, and 1983b) analyzed a number of dynamic models of reputation building. In Shapiro's analysis (1983b), for example, the firms producing high-quality goods first charge the low-quality price, allowing consumers to discover the quality without risk, and later charge and receive the incentive prices.

Performance uncertainty. In this article, we extend Telser's and Klein and Leffler's work in a different direction. In particular, we are concerned with investigating the nature of the self-enforcing, repeat-purchase equilibrium in the face of performance uncertainty. This uncertainty arises when a seller has an imperfect ability to control and measure the quality of the goods or services he

[1] The concept of reputation is only relevant where prepurchase information is imperfect. Where product attributes are perfectly observable, a firm's past actions are irrelevant to consumers. (See Kreps and Wilson (1982a) and Shapiro (1983b.)

produces. For example, a manufacturer of light bulbs cannot know with certainty how long a particular bulb will last: only the failure rate of a batch can be controlled. Similarly, an investment banker's estimate of the performance of a new security is necessarily probabilistic, and thoughtful consultants know that their advice has some chance of causing harm rather than good. In this article, we operationalize performance uncertainty by assigning a fixed level of process reliability, g, to a technology; each unit produced with the technology has an independent probability, $1 - g$, of being defective.

In the absence of performance uncertainty, the receipt of only one bad product would be sufficient to brand a producer as a low-quality supplier. But, the existence of performance uncertainty complicates the problem of creating and maintaining a reputation as a highquality producer. If the recipient of a bad product gives the seller the benefit of the doubt, profit-maximizing sellers will always cheat. If, on the other hand, buyers who receive bad products assume the worst and boycott, then sellers will not cheat, making the buyers' pessimism unwarranted.

Allen and Faulhaber (1988) studied the impact of a form of performance uncertainty, termed quality noise, on reputational equilibria. In their model, a monopolist makes a hidden choice between "good" and "bad" machines, where the better machine has a higher probability of producing a valued product each period. There are two periods, and the monopolist adjusts his second-period prices according to the quality delivered in the first period. Allen and Faulhaber showed that there is no pure-strategy equilibrium involving consistent conjectures and that an equilibrium in which the "good" machine is chosen requires buyer "skepticism." That is, a buyer must assign some weight to the chance that the seller has installed the bad machine, regardless of the incentives.

Our model and approach differ from Allen and Faulhaber's in several respects. In their model, all of the units produced in a period are identical; the performance uncertainty applies to the production process itself rather than the individual unit produced. Additionally, the producer's incentive to choose the "good" machine is derived from the lower secondperiod prices that accompany the delivery of a defective batch of first-period products. [2] But, the price adjustment mechanism is not compatible with the existence of performance uncertainty; to combine them, one must assume that buyers who receive products of different (random) quality are offered different second-period prices. This strenuous form of price discrimination is not only unrealistic, it also leaves the seller vulnerable to opportunistic misrepresentations of product quality. Consequently, our model couples the existence of performance uncertainty with a repeat-purchase mechanism, rather than a price adjustment mechanism. This structure generates high-quality equilibria without requiring the introduction of "bad" types of sellers and permits the analysis of competition.

Some features of our assumption of performance uncertainty are similar to those of Rogerson's (1983) assumption that consumers make errors in their evaluations of product quality. In Rogerson's model, buyers switch suppliers if the cumulative fraction of lowquality purchases rises above some critical level. He showed that this switching rule ensures that high-quality firms will be larger than low-quality firms. Our approach is different; rather than assume producer heterogeneity, we deduce the choice of strategy by a rational seller.

The article is organized as follows. In Section 2, we introduce the two-period, repeatpurchase model with performance uncertainty, and in Section 3, we analyze the strategies followed by the buyers and sellers when the price is set exogenously. Sections 4 and 5 present analyses of the mixed-strategy equilibria which occur under conditions of monopoly and competition. In Section 6, we explore the consequences of consumer switching. Our conclusions are presented in Section 7.

[2] In effect, the price adjustment is a form of limited quality insurance.

The Two-period Model

Firms are identical and can make a hidden selection between a low-quality and a highquality production technology. The low-quality technology produces uniformly "bad" products at a unit cost of C_L. Buyers are homogeneous in their tastes, and each is willing to pay up to V_B for one unit of the bad product. The high-quality process is capable of making products of more value to consumers, but it is not dependable. Each unit produced at a cost of C_H has an independent probability, $g < 1$, of being "good," or having a consumer value of $V_G > V_B$, and a corresponding probability, $1 - g$, of being "bad," or having a value of V_B. Neither producers nor buyers can distinguish between good and bad products without consuming (or destructively testing) them.

Buyers and producers are risk neutral. The expected value to a consumer of a product from the high-quality process is

$$V_H = V_B + g\left(V_G - V_B\right).$$

To simplify the analysis, we assume that $V_B < C_L$, preventing the announced production and sale of "bad" products. By contrast, $C_H < V_H$, so there may be a market for goods produced with the high-quality process. However, because $C_L < C_H$ and because consumers cannot observe a firm's choice of technology, a firm has an incentive to cheat by installing the less expensive low-quality process and then falsely claiming it is using the high-quality process.

There are two periods, and each of N consumers buys at most one unit in each period. Both producers and consumers have the same positive discount factor, $\beta < 1$. Consumers face two decisions: whether to make an initial purchase and whether to repurchase in the second period. We assume that repurchase decisions are sequentially rational and are made using pure private information; that is, consumers do not share information about product quality with each other. [3] Without this assumption, buyers might pool their experience information and arrive at a reliable estimate of the seller's technology choice. In such a case, performance uncertainty would have no bite, and the model is structurally equivalent to that of Klein and Leffler (1981) and its derivatives. Thus, performance uncertainty is only of interest when it is coupled with private quality information. [4]

The problem of seller opportunism disappears if unambiguous, easily enforceable product warranties are available. However, real warranties [5] are costly to enforce and do not begin to cover the myriad characteristics that constitute "quality." Chief among the factors which limit the efficacy of warranties are the adverse selection of customers, moral hazard with respect to buyer behavior and to the reporting of quality failures, difficulty in the measurement of product characteristics, and enforcement costs. Thus, in this article, we regard the issues of performance uncertainty and quality hazard to apply to those aspects of quality which cannot be efficiently insured via warranties. [6] A restaurant, for example,

can warrant that a meal will be free of toxins, but not that it will please a discriminating

[3] Consumers might not share information because it is too costly to search for other consumers of the product or because the reports of other consumers are unreliable. That is, "quality" might depend upon the difficult-tocommunicate characteristics of products and unobservable private tastes, or consumers might have incentives to misrepresent their experiences. [4] Shared learning assumptions can have unexpected but crucial implications. For example, with shared learning, a firm can signal its quality at a very low cost by providing a single sample to a single consumer. Alternatively, consumers who enter the market late can freeload by benefitting from the experience of earlier consumers, without incurring the costs of product sampling themselves. [5] The Magnuson-Moss Warranty and Federal Trade Commission Act of 1978 requires that product warranties be provided with all consumer products sold in the U.S. for more than fifteen dollars. [6] Note that warranties are invariably ruled out in the literature on reputation and self-enforcing agreements, but in models in which quality is unambiguously discerned

palate. Similarly, a software producer can warrant a program's accuracy, but not its degree of "user friendliness."

In summary, buyers are price takers and act to maximize their expected surplus. Sellers act to minimize their expected profits. The game structure and all of the cost and value parameters are common knowledge. The information asymmetries are that only the sellers know their technology choices, that product quality cannot be determined without consumption, and that consumers do not share information about their personal quality experiences.

This two-period model offers substantial advantages in modelling simplicity and in the clarity of results. However, with only two periods, the second period is the last. Consequently, some additional assumptions are necessary to avoid unravelling the model. Specifically, both the technology and the price must be set once and for all at the outset of the game.

If the technology is flexible, a producer will always secretly choose the low-quality process in the last period. Buyers will predict this choice, and the market will fail. Lacking any continuation, the first-period game also unravels. To permit a high-quality equilibrium to exist, there must be a mechanism that allows rational buyers to believe that a producer's past performance is a guide to his future performance. Our approach is to assume that technological choices involve substantial irrecoverable product-specific investments, so that firms do not change technologies at the end of the first period.

The situation regarding the price is more complex. In the model studied here, the seller's honesty is induced by conditional second-period boycotts. For these boycotts to be sequentially rational, second-period prices must be such that the recipients of bad products will indeed boycott. However, if the seller can adjust his second-period prices after firstperiod consumption has taken place, then the equilibrium is upset: pricing to eliminate boycotting also eliminates the incentive structure which induces honesty. As Allen and Faulhaber (1988) have shown, the price adjustment mechanism, by itself, does not provide a sufficient penalty to curb a seller's opportunism.

One solution to this problem is to adopt Allen and Faulhaber's (1988) assumption that some fraction of sellers belong to a "bad" type, which never produces a good product, regardless of the incentives. Given this assumption, the Bayesian downward revision in expected product value that accompanies the receipt of a bad product can be sharp enough to curb the sellers' opportunism (among the "normal" sellers). In the model studied here, however, all sellers are equally free to choose their production policies-there are no "types." Therefore, only two approaches remain to obtaining a market equilibrium: (1) postulating buyers' beliefs in such a way that a second-period price reduction from the preannounced level is taken to be a signal that the seller uses a low-quality process, [7] or (2) permitting forward price commitments. The buyers' beliefs that are required in the first approach can be supported (are "credible") because dishonest sellers have somewhat greater incentives to cut second-period prices. But, this approach has several drawbacks: it permits a continuum of equilibria and explicitly introduces the kind of arbitrary beliefs that we labor to exclude. By contrast, the price commitment assumption yields a unique market equilibrium without the introduction of types or arbitrary beliefs.

The assumption that price commitments are possible is best thought of as an agreement to sell a second product at a prearranged price if the customer is satisfied with the first. Forward price commitments can be achieved through investment in publicity, (making subsequent price changes very costly), through explicit trial period contracts, and through cancellable supply contracts, as are commonly used in equipment and property rental and

and the information is shared costlessly among many consumers, we find this assumption less than fully compelling. However, the existence of performance uncertainty, coupled with the assumption that each buyer's experience of quality is private (unverifiable) information, does rule out the use of quality warranties by the seller.

leasing, hotels, labor supply contracts, professional services, and in the provision of higher education. Government regulation can produce what are, in effect, committed prices, as can sequential contracting on a cost-plus basis. In addition, sellers can take advantage of markets with customary prices (e.g., theatrical motion pictures, magazines) in order to limit price flexibility.

The Exogenous Price Game

This section presents an analysis of the situation in which the price is set exogenously by some outside party or process. Because this game is a proper subgame of the monopoly and competition games, the results obtained are directly useful in Sections 4 and 5. Furthermore, the exogenous price game is of interest in its own right; such a situation might correspond to a regulated industry, certain forms of government contracting, or intraorganizational sourcing arrangements with administered pricing.

The game tree is shown in Figure 1. The sellers decide whether or not to produce and then, announce the price. If they decide to produce, they make the (hidden) decision whether to install the high-quality technology (High) or the low-quality technology (Low). The producer's strategy is described by $h \in [0, 1]$, the probability that the high-quality process is chosen. Because sellers always claim to be using the high-quality process, we refer to h as the seller's "honesty." Firms that produce always do so for both periods; strategies of producing in only one period are dominated by producing in both periods whenever producing at all dominates Out.

If the seller chooses to produce, consumers then decide between purchasing or doing without (Notbuy). If a first-period purchase is made, the product is either good or bad. The buyer must then choose between repurchasing (Rebuy) and forgoing the second-period offering (Boycott). The buyer's strategy is described by $r \in [0, 1]$, the probability of repurchase. We refer to r as the buyer's "trust" in the seller.

One valid Nash equilibrium to this game is market failure: the seller chooses Out, and buyers choose Notbuy. However, our main interest is in equilibria that support exchange.

We define a market equilibrium as a sequential equilibrium [8] in the game in which the sellers produce, choosing between High and Low, and in which consumers buy, choosing between Rebuy and Boycott in the second period. Thus, a market equilibrium in the exogenous price game is a strategy pair, $(\ h; r\ )$, such that each strategy is a best response to the other and such that the equilibrium provides nonnegative seller profits and a nonnegative consumer surplus.

Sellers' reactions. The scaling is such that the seller's strategy depends only on his profits per unit and is insensitive to the number of buyers. The total expected profit per first-period unit, Π, [9] for a seller using honesty h in response to price p and trust r may be written as

$$\Pi(p, h, r) = h[1 + g\beta + r\beta(1 - g)]\,(p - C_H) + (1 - h)(1 + r\beta)\,(p - C_L)\,.$$

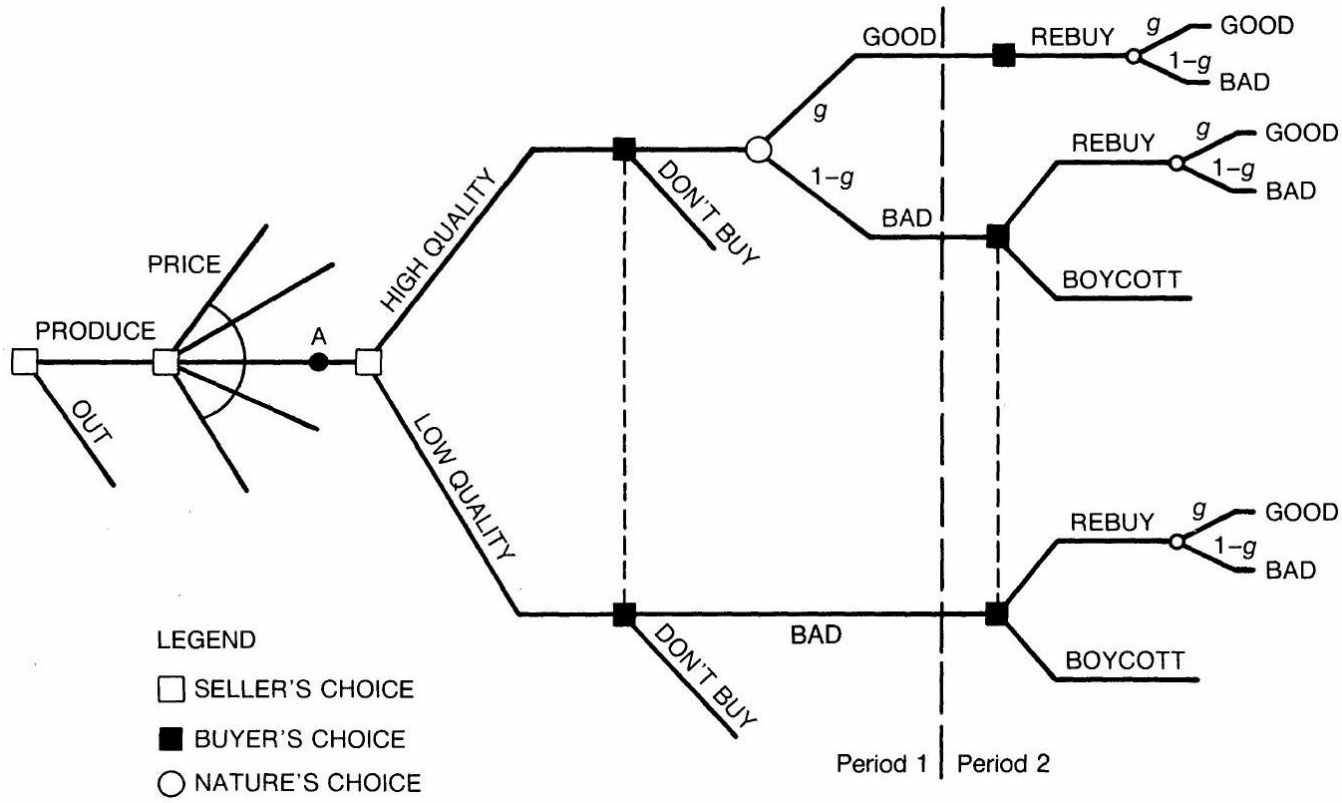
Now, consider the seller's choice of h. Define $\Pi_\Delta(p, r) = \Pi(p, 1, r) - \Pi(p, 0, r)$, the gain per unit from playing High rather than Low. Additionally, define the price

$$P_0 = C_H + \frac{(C_H - C_L)}{\beta g}$$

so that Π_Δ may be written as

[7] For example, consider an employer's attitude towards an employee who responds to a negative performance appraisal by offering to accept lower wages.

FIGURE 1 EXTENSIVE FORM OF ENDOGENEOUS-PRICE GAME TREE



$$\Pi_\Delta(p,r) = (1-r)g\beta\,(p-P_0) - r(1+\beta)\,(C_H - C_L)\,.$$

If $\Pi_\Delta > 0$, the seller installs the high-quality process ($h = 1$). If $\Pi_\Delta < 0$, the low-quality process is chosen ($h = 0$), and if $\Pi_\Delta = 0$, the seller may select any $h \in [0,1]$.

Observe that if buyers are trusting ($r = 1$), then $\Pi_\Delta(p,1) < 0$, and a seller's best response is to cheat and play Low. Even if buyers are completely skeptical ($r = 0$), sellers will still play Low if $p < P_0$. The problem is that at prices below P_0, the profits lost through second-period boycotts are too small to offset the gain from cheating. Thus, boycott strategies are only capable of inducing seller honesty at prices of P_0 or more. P_0 is greater than the unit cost and is equivalent to the Klein-Leffler "quality-assuring price" discussed in Section 1. Thus, a necessary condition for a market equilibrium to exist is that $p \geq P_0$. It follows that the (second-period) price will be above the cost in any exchange.

If $p > V_H$, the expected consumer surplus from the high-quality process is negative, and customers play Notbuy. Thus, a second necessary condition for a market equilibrium to exist is that $p \leq V_H$. Define a feasible price as one which falls in the interval $[P_0, V_H]$. To ensure the presence of a range of feasible prices, we require that $P_0 < V_H$. [10]

The condition $P_0 < V_H$ implies that $V_G - V_B > (P_0 - V_B)/g$. If g is made sufficiently small, this constraint will be violated. Thus, there is a lower bound on the level of process reliability. Combining the above condition with (3) yields the condition $g > g_0$, where

$$g_0 = \frac{C_H - V_B}{2\,(V_G - V_B)}\left[1 + \sqrt{1 + \frac{4\,(V_G - V_B)\,(C_H - C_L)}{\beta\,(C_H - V_B)^2}}\right]$$

In what follows, we assume $g > g_0$. Returning to the investigation of Π_Δ, for $p \in [P_0, V_H]$, the seller chooses Low if $r = 1$ and chooses High if $r = 0$. Because Π_Δ is linear and decreasing in r, there is a unique $\hat{r} \in [0, 1]$ that makes the seller indifferent between High and Low, where

$$\hat{r}(p) = \frac{p - P_0}{p - C_H + \beta\,(P_0 - C_H)}$$

It follows that $\Pi_\Delta(p, r) \geqq 0$ if and only if $\hat{r}(p) \geqq r$, which permits the seller's best response, $h^*(p, r)$, to a buyer's strategy, r, to be characterized as follows:

$$\begin{array}{lll} & h^* = 1 & \text{if} \quad r < \hat{r}(p), \\ & h^* \in [0,1] & \text{if} \quad r = \hat{r}(p), \\ \text{and} & h^* = 0 & \text{if} \quad r > \hat{r}(p). \end{array}$$

Buyers' reactions. The expected consumer surplus per unit, S, is

$$S(p, h, r) = h[1 + g\beta + r\beta(1 - g)]\,(V_H - p) + (1 - h)(1 + r\beta)\,(V_B - p)\,.$$

We require that buyer's behavior be sequentially rational. In particular, given the private quality information, $Q \in \{G, B\}$, we require that buyers use Q, the seller's strategy h, and Bayes' rule to generate $h^Q(h)$, a posterior estimate of the seller's honesty, and then act to maximize their expected surplus given that estimate. If the first-period product is good, the buyer knows that the high-quality process is in use, or that $h^G(h) = 1$. If the first-period product is bad, the buyer uses Bayes' rule to obtain

$$h^B(h) = \frac{h(1 - g)}{1 - hg}.$$

A buyer expects a second-period surplus of $S^Q(p, h)$, where $S^G(p, h) = V_H - p$ and

$$S^B(p, h) = V_B + \frac{g(1 - g)h}{1 - hg}\,(V_G - V_B) - p$$

If the first-period product is good, then buyers know that the high-quality process is in use and are willing to pay up to V_H for the second-period product. Thus, buyers receiving good products always play Rebuy in the second period. [11]

The behavior of a recipient of a bad first-period product depends upon the sign of $S^B(p, h)$. Using (12) gives $S^B(p, 0) = V_B - p < 0$ and $S^B(p, 1) = V_H - p \geq 0$ for feasible prices. These end conditions, along with the fact that S^B is strictly increasing in h, ensure the existence of a unique $\hat{h}(p) \in [0, 1]$ that makes the recipient of a bad product indifferent between Boycott and Rebuy in the second period, where

$$\hat{h}(p) = \frac{p - V_B}{p - V_B + (1 - g)\,(V_H - p)}$$

It follows that $S^B(p, h) \gtreqless 0$ if and only if $h \geqq \hat{h}(p)$, which permits this characterization of a buyer's best-response, $r^*(p, h)$:

[8] Kreps and Wilson (1982a). [9] Here, and hereafter, references to money flows "per unit" are taken to be the expected present profits, surplus, or costs associated with each unit sold in the first period. [10] Summarizing the assumptions and requirements governing the parameter ordering, we have that $V_B < C_L < C_H < P_0 < V_H$.

$$\begin{array}{rll} & r^* = 0 & \text{if} \quad h < \hat{h}(p), \\ & r^* \in [0,1] & \text{if} \quad h = \hat{h}(p), \\ \text{and} & r^* = 1 & \text{if} \quad h > \hat{h}(p). \end{array}$$

Given a feasible price, p, and a proposed market equilibrium, $(h; r), \pi(p, h, r) > 0$ is ensured because $p \geq P_0 > C_H$. However, there are values of h below which $S < 0$. Since $r = 0$ when $h < \hat{h}$, the value of this lower limit on honesty, h_0, is obtained by solving $S(p, h, 0) = 0$, so

$$h_0 = \frac{p - V_B}{p - V_B + (1 + g\beta)(V_H - p)}.$$

Comparing (13) with (17) confirms that $h_0 \leq \hat{h}$ for feasible prices. Proposition 1. There exists a market equilibrium if and only if $p \in [P_0, V_H]$. For every feasible price p, there is a market equilibrium ($\hat{h}(p); \hat{r}(p)$). If $P_0 < p < V_H$, this equilibrium is unique. If $p = V_H$, the set of market equilibria is $\{(1; r) : r \in [0, \hat{r}(V_H)]\}$. If $p = P_0$, the set of market equilibria is $\left\{(h; 0) : h \in \left[h_0(P_0), \hat{h}(P_0)\right]\right\}$.

Proof. If $p > V_H$, the buyer's surplus is negative, even if $h = 1$, and if $p < P_0$, the seller's best strategy is always Low. It follows that there cannot be exchange at these prices. So, we restrict our attention to feasible prices. Using (8) and (15), the pair ($\hat{h}$; $\hat{r}$) satisfies $\hat{h} \in h^*(p, \hat{r})$ and $\hat{r} \in r^*(p, \hat{h})$. Because $\hat{h} > h_0$ and $p > C_H$, both profits and the surplus are nonnegative. Thus, $(\hat{h}; \hat{r})$ is a market equilibrium.

Now, let us restrict our attention to a price that is interior to the feasible interval: $P_0 < p < V_H$. From (13) and (6), we have that $0 < \hat{h} < 1$ and $0 < \hat{r} < 1$ for an interior price. Suppose ($h; r$) is an equilibrium and $h < \hat{h}$. Then, by (14), a buyer's best response is $r = 0$. But, by (7), the seller's best response to $r = 0$ is $h = 1$, contradicting the supposition. Suppose, instead, that $h > \hat{h}$. Then, by (16), a buyer's best response is $r = 1$. But, by (9), the seller's best response to $r = 1$ is $h = 0$, contradicting the supposition. Similar arguments with regard to $r \neq \hat{r}$ establish the uniqueness of the equilibrium.

If $p = P_0$, then $\hat{r} = 0$, and the buyer's response of $r = 0$ is consistent with

$$h_0(P_0) \leq h \leq \hat{h}(P_0).$$

Hence, market equilibria exist at $p = P_0$ with $r = 0$ and $h \in \left[h_0(P_0), \hat{h}(P_0)\right]$. If $p = V_H$, then the seller's strategy of $h = 1$ is consistent with $r \in [0, \hat{r}(V_H)]$. Q.E.D.

Corollary 1. A market equilibrium in pure strategies exists if and only if $p = V_H$. It is unique and entails the seller playing High and the buyer playing Boycott.

Corollary 1 clearly rules out the classic Klein-Leffler (1981) equilibrium, in which sellers play High, buyers play Boycott, and the price is P_0. If $g = 1$, the buyer never receives bad products from an honest producer. But, if there is performance uncertainty, buyers will receive bad products even when sellers choose High. Unlike Klein and Leffler's consumers, these buyers will need to boycott against sellers known to be honest. However, sequentially rational buyers will not carry out such boycotts unless the price rises to their limit price, V_H, ensuring indifference. Thus, if players are restricted to pure strategies, there can be no distinction between competition, monopoly, or any other process for setting price-the only possible equilibrium price is V_H.

In the mixed-strategy equilibria, consumers choose $\hat{r}$ to make producers indifferent between playing High or Low, and producers select $\hat{h}$ to ensure that a buyer who receives a bad product

[11] A strategy of sometimes boycotting good products when $p = V_H$ acts to reduce the seller's incentive to be honest.

makes a posterior estimate of honesty, $h^B[\hat{h}(p)]$, that yields $S^B(p,h) = 0$. The seller's equilibrium strategy makes credible the buyer's threat to boycott bad products.

Proposition 1 establishes that $(\hat{h};\hat{r})$ is the unique equilibrium if prices are in the interior of the feasible interval and that at the endpoints of the feasible interval, there are multiple equilibria. In the following comparative statics discussion, we shall focus on those endpoint equilibria which are the limits of the continuous functions $\hat{r}(p)$ and $\hat{h}(p)$. These equilibria are reasonable focal points because of their continuity, because they are Pareto superior to the other endpoint equilibria, and because they become the unique endpoint equilibria once the selection of the price is made endogenous (Propositions 3 and 4).

Because $\Pi_\Delta(p,\hat{r}) = 0$, the value of $\Pi(p,h,\hat{r})$ is independent of h (seller indifference). Consequently, the seller's equilibrium profit, $\hat{\Pi}$, can be obtained from (2) by selecting h to be any convenient quantity in the range $[0,1]$. Using $h = 0$ yields

$$\hat{\Pi}(p) = \Pi(p,0,\hat{r}) = (1+\hat{r}\beta)\,(p - C_L)\,.$$

Similarly, $S(p,\hat{h},r)$ is independent of r. Setting $r = 1$ yields the equilibrium surplus

$$\hat{S}(p) = S(p,\hat{h},1) = (1+\beta)\left(V_B + g\hat{h}\,(V_G - V_B) - p\right) = (1+\beta)\,(V_H - p)\,g\hat{h}.$$

Figure 2 illustrates typical forms of $\hat{h}(p)$ and $\hat{r}(p)$. Observation 1. The equilibrium values of honesty, $\hat{h}$, trust, $\hat{r}$, and seller's profit, $\hat{\Pi}$, are increasing in price.

Note that the producers are fully trustworthy only if the price is at its upper limit ($p = V_H$), but are never wholly dishonest. Buyers are completely skeptical only if the price is at its lower limit ($p = P_0$), but are never wholly trusting. Thus, consumers have some positive probability of boycotting producers after the purchase of bad products, even when they are certain that producers are using the high-quality process. It is this "rational skepticism" on the part of consumers that makes the producer strategy $\hat{h}$ stable. It is only because consumers play Boycott some of the time that firms continue to play High rather than Low. Observation 2. The equilibrium values of trust, $\hat{r}$, seller's profit, $\hat{\Pi}$, and consumer surplus, $\hat{S}$, are increasing in the process reliability, g.

If the value of V_H is held constant and changes in g are made to change the value of V_G, then $\hat{h}$ is increasing in g. If, instead, V_G is held constant, then V_H increases in g, and $\hat{h}$ may increase or decrease in g, depending upon the value of p. For example, if $p = V_H$, then $\hat{h} = 1$, and an increase in g yields $p < V_H$ and $\hat{h} < 1$.

Define the total surplus (producer and consumer) per unit as $\hat{T} = \hat{S} + \hat{\Pi}$. The following result may be unexpected.

Proposition 2. The equilibrium total surplus is increasing in price. Proof. At the equilibrium, $\partial S/\partial r = \partial\Pi/\partial h = 0$, and it is clear from (10) and (2) that $\partial S/\partial p + \partial\Pi/\partial p = 0$. Hence,

$$\frac{\partial\hat{T}}{dp} = \frac{\partial\Pi}{\partial r}\left[\frac{d\hat{r}}{dp}\right] + \frac{\partial S}{\partial h}\left[\frac{d\hat{h}}{dp}\right] > 0$$

This result occurs because a price increase has two distinct effects: it transfers wealth from the consumers to the producers, which has no effect on the total surplus, and it increases honesty, $\hat{h}$, and trust, $\hat{r}$, which, other things being equal, increases the consumer surplus and profits. Of course, much of the sharpness of this result comes from the underlying assumption that the consumers have identical reservation prices, so price increases that leave the price short of the reservation price do not influence demand.

The Monopoly Equilibrium

A monopoly market equilibrium is a triple, ($p, h; r$), where the semicolon separates the seller's choices from the buyers'.

A monopolist who decides to produce begins the game by announcing the price p. Were the price exogenous, it is clear from Observation 1 that the seller's profits would be largest if $p = V_H$; this is the price the seller most prefers in the exogenous price game. Therefore, it seems intuitively compelling that a monopolist would choose $p = V_H$. This is indeed the case if the game tree is constructed as shown in Figure 1. Consider node A in that tree. With the monopolist's price announcement already made, the ensuing subgame is the exogenous price game analyzed in Section 3. In particular, note that the price cannot be a signal because the buyer has no uncertainty about which node has been reached. There are no preceding hidden moves by either player or by nature, so there can be no beliefs about such moves to be conditioned on the price announcement. Consequently, the choice of p defines the subgame to be played, and the monopolist will use this power to select the subgame that provides the greatest expected profits.

However, if the game tree is trivially transformed by interchanging the order of the seller's price and technology choices, [12] a continuum of sequential equilibria arise, since the definition of a sequential equilibrium permits buyers' off-the-equilibrium-path beliefs about the seller's technology to be conditioned on the price announcement. Consider, for example, the following sequential equilibrium: The seller sets $p = \xi$, where $P_0 < \xi < V_H$. The seller's honesty is $\hat{h}(\xi)$, and the buyers' trust is $\hat{r}(\xi)$. Buyers believe $h = 0$ if the price announcement is $p > \xi$. Here, the buyers' off-the-equilibrium-path beliefs act to enforce the equilibrium.

There are clearly a profusion of such sequential equilibria in this game tree. We view these equilibria as "unreasonable" for two reasons. First, there is no motivation for one particular arbitrary belief to be common knowledge among a pool of anonymous buyers. Second, the seller's incentive to choose High rather than Low is influenced by the anticipated price choice (h increases in p), but once the technology is chosen, there is no motivation to attach additional signalling value to it; both high- and low-quality producers will produce at any feasible price, and both want the price to be as high as possible.

It is possible to rule out those equilibria in which price is less than V_H by using refinements of the concept of a sequential equilibrium in the spirit of those advanced by Grossman and Perry (1986). Their argument is that beliefs assigned to off-equilibrium information sets must be credible; that is, they should be among the set of internally consistent beliefs for the player were that information set actually reached. A brief sketch of the refinement we have in mind follows. Suppose that the seller were to defect from the proposed equilibrium (above) and announce a price of $p = V_H$. First, the buyers should realize that they are expected to buy, or else the defection would be irrational. But, if the buyers buy, the only consistent belief about seller honesty, at this price, is $h = 1$. To see why, note that the seller

anticipates that the defection induces introspection and purchase. (Why else defect?) Given that purchases are made, the buyers' beliefs that $h < \hat{h}$ lead to a full boycott of bad products ($r = 0$), which should have induced the anticipating seller to select $h = 1$, a choice inconsistent with the espoused belief. Hence, it is not credible for a consumer who buys to insist on holding beliefs other than $h = 1$. It is also not reasonable for a consumer who realizes this not to buy. The seller is better off at the new equilibrium, which rationalizes the defection, and undermines the original proposed equilibrium. This revision rule leads to the unique monopoly equilibrium developed below

[12] Although these decisions are best thought of as simultaneous, the formalism of the extensive form requires that one be placed before the other.

(Proposition 3).

Fortunately, there is a less fragile and more straightforward way to eliminate bizarre equilibria in this game. In their discussion of strategic stability, Kohlberg and Mertens (1986) pointed out that sequential equilibria can depend upon arbitrary details in the construction of the game tree. In particular, they argued that reasonable equilibria should, among other things, satisfy an invariance condition-they should survive inessential transformations of the tree. One of six inessential transformations is the interchange of simultaneous moves. Define a robust equilibrium as a sequential equilibrium which survives interchanges of the order of the seller's price and technology decisions. [13] The monopoly equilibrium proposed above ($\xi < V_H$) is not a sequential equilibrium in the price-first extensive form; therefore, it is not robust.

The link between robustness and the refinement argument sketched above is instructive. Recall that sequential equilibria are a subset of all the Nash equilibria and that Nash equilibria are independent of the extensive form representation. (They follow from the game's normal form.) Consider a sequential equilibrium in the technology-first tree in which arbitrary beliefs lead to an equilibrium price of $\xi < V_H$. Now, suppress the issue of beliefs, and examine the corresponding Nash equilibrium. In it, buyers follow the strategy "Notbuy if price is more than ξ." In the price-first tree, such strategies do not constitute a Nash equilibrium in the subgame and, therefore, are ruled out. But, in the technology-first tree, they are valid sequential equilibria supported by appropriate beliefs. Thus, every equilibrium that is based on incredible beliefs in the technology-first tree corresponds to an equilibrium that is based on incredible threats in the price-first tree.

In what follows, we shall require that market equilibria be robust. Since it appears that the set of sequential equilibria in the price-first tree is a subset of those in the technologyfirst tree, we shall work with the price-first tree. Then, to establish that an equilibrium in the price-first tree is robust, it suffices to find some set of beliefs that support the same market equilibrium in the technology-first tree.

With these preparations, we can prove that monopolists will charge the monopoly price-and be completely honest.

Proposition 3. The unique monopoly market equilibrium is $(V_H, 1; \hat{r}\,(V_H))$. Proof. We shall work with the price-first game tree. The monpolist will select p so as to inaugurate the subgame which provides the greatest profits. Because $\hat{\Pi}$ is increasing, the equilibrium ($V_H, 1; \hat{r}\,(V_H)$) provides the greatest profits. To establish uniqueness, consider a potential equilibrium, ($V_H, 1; \rho$), where $\rho < \hat{r}\,(V_H)$, yielding producer profits of

$$\hat{\Pi}_\rho = [1 + \beta\rho + g\beta(1-\rho)]\,(V_H - C_H)\,.$$

For any such ρ, there is a very small price reduction, ϵ, which would yield superior profits. To see this, note that a buyer's best response to a price $V_H - \epsilon$ is $\hat{r}\,(V_H - \epsilon)$ and that $\hat{r}\,(V_H - \epsilon) > \rho$ for ϵ sufficiently small. Thus, the seller can come arbitrarily close to

$\hat{\Pi}\,(V_H) > \hat{\Pi}_\rho$ by choosing ϵ sufficiently small. It then follows that $(V_H, 1; \rho)$ is not an equilibrium.

To establish robustness, assign the beliefs $h_b = \ddot{h}(p)$ in the technology-first tree. These are the "beliefs" a buyer would hold if the price decision had come first. Therefore, they support the equilibrium ($V_H, 1; \hat{r}\,(V_H)$). Hence, it is robust. Q.E.D.

Notice that this monopoly equilibrium is not identical to the pure-strategy equilibrium of Corollary 1. In both cases $p = V_H$ and $h = 1$, but the admission of mixed strategies has raised the equi-

[13] We are unaware of any other application in which this invariance condition has been used to prune equilibria. The condition is only one of many that Kohlberg and Mertens (1986) imposed to define stability. Robustness is therefore much less stringent than stability; robust equilibria are not necessarily stable equilibria.

librium level of trust, $\hat{r}$, above zero. The intuition is straightforward: full boycotting is a sufficient, but unnecessarily harsh, method of inducing producer honesty.

If the assumption that price commitments are possible were relaxed, this equilibrium would collapse. In the second period, even an honest seller has an incentive to reduce the preannounced price by a small amount if, by doing so, all buyers are induced to rebuy. Since this is true for any proposed monopoly price and since sellers will always choose Low if all buyers rebuy, there can be no market equilibrium with flexible second-period pricing. It is possible to obtain a market equilibrium with flexible prices by assigning off-path beliefs to buyers such that they boycott a seller who reduces the preannounced price. But, such a specification would introduce a continuum of other signalling equilibria and would violate the spirit of the rational-buyer model we have constructed.

Competitive Equilibria

In this section, we extend our analysis to competitive market equilibria by allowing buyers to select which of many two-period games they wish to play. We assume that no new sellers appear in the second period and that buyers are locked-in to their suppliers. [14] Thus, the buyers' second-period choices continue to be Rebuy and Boycott. In Section 6 we enlarge the model to include new sellers and buyer switching.

We envision a competitive process in which the buyers patronize the seller or sellers who offer the most favorable prices. In equilibrium, no firm should find it profitable to lower (or raise) its price, and all firms must earn zero net economic profits. In a commodity market, this process would drive the price down to the cost. However, in the present setting, the price plays the additional role of structuring the seller's incentive to be honest. Thus, from a buyer's viewpoint, the most favorable prices are not the lowest prices. A similar situation has been studied in labor markets; if the productivity of labor rises with the wage rate, the efficiency wage, which minimizes the costs per unit of labor, is not the lowest possible wage. [15]

Given a price announcement has been made, both the seller and consumers know they are at node A in Figure 1 and, as in the case of monopoly, can see that the ensuing subgame is the exogenous price game. The buyers can calculate the expected surplus, $\hat{S}(p)$, obtained in the equilibrium of the proposed subgame. Given that several price announcements have been made, the buyers select the one that yields the largest $\hat{S}(p)$, randomizing among the group of sellers offering the best terms. Thus, the competitive price, γ, is that price which delivers the greatest consumer surplus and is also consistent with nonnegative producer profits. However, price increases enhance seller honesty, so in some situations, the consumer surplus, $\hat{S}$, increases with price. Consequently, the competitive price, γ, is not necessarily P_0, the lowest feasible price.

Proposition 4. Define

$$\theta = V_B + (V_G - V_B)\left[\sqrt{1-g} - (1-g)\right]$$

Then, the competitive price is $\gamma = \max(P_0, \theta)$, and the unique, competitive equilibrium is $(\gamma, \hat{h}(\gamma); \hat{r}(\gamma))$.
Proof. The function $\hat{S}(p)$ is defined for $p \in [V_B, V_H]$ and is strictly concave on that interval. The price θ is the solution to $d\hat{S}/dp = 0$, and $\theta \in [V_B, V_H]$; therefore, $\theta = \operatorname{argmax} \hat{S}(p)$. If $\theta \geq P_0$, it is feasible and, therefore, the competitive price; the resulting unique equilibrium is $(\theta, \hat{h}(\theta); \hat{r}(\theta))$. If

[14] If, instead of selecting Rebuy, disgruntled buyers switch to other, randomly chosen sellers in the second period, each seller expects to receive as many new buyers as he loses, so the equilibrium is essentially unchanged. [15] See the review by Stiglitz (1987), in which a variety of similar phenomena are discussed.

$\theta < P_0$, it is infeasible, and $\gamma = P_0$, as $\hat{S}$ is decreasing in p everywhere above θ. Uniqueness and robustness follow from arguments paralleling those used in Proposition 3. Q.E.D.

Figure 3 illustrates a situation in which $\theta > P_0$. At the maximum,

$$\hat{S}(\theta) = (1+\beta)\,(V_G - V_B)\,(1-\sqrt{1-g})^2.$$

The customers are totally skeptical if $\gamma = P_0$ and are more trusting if $\gamma > P_0$. The following observation is an immediate consequence of Proposition 4.

Observation 3. At the competitive equilibrium, firms are not completely honest. The classic Klein-Leffler equilibrium is $(P_0, 1; 0)$; the price is at its lowest feasible level, producers choose the high-quality process, and consumers uniformly boycott the producer if a bad product is received. If $g \to 1$, then $\hat{h}\,(P_0) \to 1$, showing that our competitive model reproduces their result. However, once performance uncertainty is introduced, the producers' trustworthiness lapses.

The following point is a direct consequence of Proposition 2. Observation 4. The total (consumer plus producer) surplus is lower under competition than it is under monopoly.

Buyers benefit from competition, but the total surplus declines because sellers are less honest as the price falls. As in Proposition 2, much of the sharpness of this result comes from the assumption that the consumers have identical reservation prices.

Rent dissipation. The competitive equilibrium does not drive the price down to the cost. From (18), firms earn the incentive rent $\hat{\Pi}(\gamma) = (1+\hat{r}(\gamma)\beta)\,(\gamma - C_L)$. To obtain the conventional competitive condition of zero expected profits, it is usually assumed that producers compete for access to these rents. In this rent-seeking rivalry, producers make nonsalvageable firm-specific investments that equal the expected value of future rents, restoring the zero-profit condition. These investments are best thought of as expenses incurred in obtaining access to the market.

Another way to recover the zero-profit condition is to posit fixed and sunk setup costs, F, for each firm. In this setting, firms enter until the expected profits from their share of the N buyers just recovers the sunk cost, F. In equilibrium, n firms enter, where n satisfies $N\hat{\Pi}(\gamma) - nF = 0$.[16]

A third way rents can be dissipated is through the offer of first-period discounts to consumers. By offering low first-period prices, producers compete to get customers to sample their products and thereby gain information about their technologies. Shapiro (1983b) identified this strategy as an investment in firm-specific consumer learning.

Offering a first-period discount, d, is equivalent to setting a price of $p-d$ in period 1 and a price of p in period 2 . A market equilibrium for this game consists of a quadruple, ($d, p, h; r$), where sellers choose d, p, and h and buyers select r.

Proposition 5. The unique competitive market equilibrium in the two-price regime is $\left(\hat{\Pi}\,(V_H)\,, V_H, 1; \hat{r}\,(V_H)\right)$

Proof. The first-period discount has no effect on the indifference functions, $\hat{r}$ and $\hat{h}$, as behavior is conditioned by the second-period price. Hence, given a price, p, and a firstperiod discount, d, the equilibrium consumer surplus is $\hat{S}(p) + d$, and the seller's profit is $\hat{\Pi}(p) - d$. For any price p, competition would force $d = \hat{\Pi}(p)$; therefore, the consumer surplus under competition is $\hat{S}(p)+\hat{\Pi}(p)$, which, by Proposition 2 , is maximized when $p = V_H$. The competitive two-price equilibrium is therefore obtained by setting the price equal to the monopoly price in the second period and offering a rebate of producer profits in the first period.

Robustness is easily established by assigning the buyers the beliefs $h_b(d,p) = \hat{h}(p)$ in the technology-first tree. Q.E.D.

Notice that the competitive two-price equilibrium provides more honesty, more trust, more total surplus, more consumer surplus, and less producer profit than the level-price competitive equilibrium. Indeed, this equilibrium, like the monopoly equilibrium, provides the maximum possible

social surplus consistent with the producers installing the highquality process.

In equilibrium, the first-period discount is $\hat{\Pi}(V_H)$, and it follows that the first-period price is $V_H - d = C_L - \hat{r}(V_H)\beta(V_H - C_L)$. Consequently, the first-period price is less than C_L; it may even be negative, representing a bonus to buyers to induce sampling.

Price and discount constraints. We have shown that competition compels seller honesty when first-period discounts are introduced. However, in many competitive situations, there are various constraints on a producer's ability to offer two-price schemes. With constraints on pricing, less efficient results might be obtained.

One type of constraint arises from government regulation. For example, in defense and related procurement procedures, cost-plus and "reasonable-profit" constraints may limit the maximum price a seller can set. Since, with competitive first-period discounting, consumers receive all of the social surplus, it should be clear from Proposition 2 that con-

straints on maximum prices make sellers less than fully honest and reduce the consumer surplus.

Another type of constraint is on the size of discounts (or lower-limits on prices). These constraints can arise through laws against "dumping" and predatory pricing. Implicit constraints can arise if consumers are not homogeneous in their tastes. Suppose, for example, that in addition to the N consumers in our model, there exists a large group of consumers who assign a zero value to the good form of the product. If large first-period discounts drive the price to be negative, representing bonuses paid for sampling, this second group of consumers will clearly accept the bonuses but will never rebuy. The existence of these consumers constrains prices to be positive.

Still another consideration that might limit the efficacy of two-price schemes in real situations is the inability of the firm to perfectly discriminate between new and repeat purchasers. Unless new customers are clearly identifiable, firms may be unable to implement price or discount schemes that assure complete honesty.

Discussing the maximum discount constraints is simpler than discussing the minimum price constraints because the former builds on relationships that have already been developed. Suppose there is a maximum allowable discount, d_u, where $d_u < \hat{\Pi}(V_H)$. (The constraint is binding.) If the price is set at V_H, the seller will pay a discount, d_u, and still have profits remaining. The following proposition establishes that competition will then force a reduction in the price. Proposition 6. If $d_u < \hat{\Pi}(V_H)$, then the competitive price is $\gamma_d = \max(\gamma, \eta_d)$, where η_d is the unique price that satisfies $\hat{\Pi}(\eta_d) = d_u$. The unique market equilibrium, $(d_u, \gamma_d, \hat{h}(\gamma_d); \hat{r}(\gamma_d))$, offers a lower price, less seller honesty, and less total surplus than the unconstrained equilibrium.

Proof. A competitive constrained equilibrium generates values of p and d which maximize the consumer surplus, $\hat{S}(p) + d$, subject to $d \leq d_u$ and $d \leq \hat{\Pi}(p)$. Since prices less than γ are Pareto inferior to γ, the solution is $\gamma_d \in [\gamma, V_H]$. If the discount constraint binds, the consumer surplus is $S_d = \hat{S}(p) + d_u$, which is decreasing in p. If the constraint does not bind, then $S_d = \hat{S}(p) + \hat{\Pi}(p)$, which is increasing in p. First, compare the case in which $\hat{\Pi}(\gamma) \geq d_u$ with that in which the constraint is binding for all prices in $[\gamma, V_H]$. Since S_d decreases throughout the range, it is clearly maximized when $\gamma_d = \gamma$. Next, consider the case in which $\hat{\Pi}(\gamma) < d_u$; since $\hat{\Pi}(V_H) > d_u$, by assumption, and $\hat{\Pi}$ is strictly increasing, there is a unique η_d satisfying $\hat{\Pi}(\eta_d) = d_u$. For prices less than η_d, S_d is increasing, and for prices greater than η_d, it is decreasing. Hence, $\gamma_d = \eta_d$ in the second case. Q.E.D.

[16] The usual problems arise if n must be an integer. A simple resolution is to allow randomization of the entry decision.

Independent Repetitions With Switching

The two-period model developed in Section 2 provides the buyer with only two secondperiod options: to repurchase or to do without the product. However, if there are many competing producers, the recipient of a bad first-period product might also turn to a new supplier in the second period. The purpose of this section is to investigate the implications of replacing Boycott (doing without) with the option of switching to a new supplier; call the new option Switch.

If some buyers search for fresh sellers in the second period, as well as in the first, sellers have the incentive to install the low-quality process, operate for one period, and then reenter the market under a different name in order to attract new first-time customers. If seller reentry costs are minimal, the existence of this option is sufficient to cause market failure. Knowing that they must be facing fly-by-night suppliers, consumers would eschew purchase. Consequently, to make the independently repeated game interesting, we assume that the sellers' reentry costs are sufficiently high to prevent sellers from presenting themselves as "new" firms in each period. Alternatively, we could assume that the technology used creates a "brand" that enables the consumers to tell whether a firm's technology has been used in prior periods. Then, customers would not purchase from a seller offering first-period trials for what they recognize to be an old "brand."

We assume that each buyer has a continuing demand for the product, wanting to consume one unit per period indefinitely. Producers exist for only two periods, with the production technology (Low or High) committed at the outset. New producers appear each period, and buyers always have the option of searching for a new producer who is just beginning production. Whenever a buyer seeks a new produer, whether at the start of the game, after Rebuy, or after Switch, there is a search or switching cost, K.

Although this model contains an indefinite number of repetitions, it is not an "infinitely repeated" game. Instead, the model embeds the isolated two-period game in a web of identical but independent games. In particular, each producer's identity (brand) lasts for only two periods. This rules out the possibility of long-term reputations and the class of subgame perfect punishment strategies that generate supergame equilibria.

Exogenous prices. We seek the stationary equilibria of this game. To characterize these equilibria, assume, for the moment, that the price p has been fixed, and isolate an arbitrary two-period element within the web of games. This two-period element is like the game of Section 3, except that each buyer faces the prospect of searching for a new supplier after making his last purchase from the current supplier. Let E be the infinite horizon net discounted surplus expected by a buyer who is about to search for a new supplier. Then, a buyer electing Switch immediately receives a payoff of E, and a buyer electing Rebuy receives $(V_H - p) + \beta E$. Thus, E is always received after the last purchase; only the timing is at issue.

Assuming, for the moment, that the termination value, E, is fixed, we can apply the analysis of Section 3 to get the following results: the feasible prices lie in the interval $[P_0, V_H - (1-\beta)E]$, the equilibrium level of trust remains $\hat{r}(p)$, and the equilibrium producer's profit remains $\hat{\Pi}(p)$. The equilibrium level of honesty is

$$\hat{h}_E(p) = \hat{h}(p + (1-\beta)E),$$

and the expected consumer surplus in equilibrium is

$$\hat{S}_E(p) = \hat{S}(p + (1-\beta)E) + E.$$

Now, recognize that stationarity requires that E be the same at every node at which search begins. In particular, at the node which led to the search for the game analyzed above, stationarity requires

that

$$\hat{S}_E(p) - K = E.$$

That is, the expected surplus from the two-period element just analyzed, less the search cost of locating that supplier, must equal the terminating value. Thus, the solution to the repeated game is obtained by imposing a fixed-point condition on a two-period game with terminating values.

The solution to (23) is a market equilibrium if $\hat{E} \geq 0$, which ensures that consumers will buy, and if $\hat{E} \leq (V_H - p)/(1-\beta)$; the value of Switch cannot exceed the value gained from buying from a stream of perfectly honest suppliers, or else even the recipients of good first-period products will play Switch in the second period.

Intuition about the structure of the solutions to (23) can be achieved by defining the transformation $x = p + (1-\beta)E$. The new variable, x, has a natural interpretation as the opportunity cost of repurchase-the amount paid plus the value forgone by electing to play Rebuy rather than Switch. Applying this transformation to (21) and (22) yields $\hat{h}_E = \hat{h}(x)$ and $\hat{S}_E = \hat{S}(x) + E$, which allows (23) to be rewritten as

$$\hat{S}(x) = K$$

Thus, the problem of finding the fixed points of (23) is equivalent to the problem of finding prices that yield a surplus of K in the two-period game of Section 3. (Refer to Figure 3.) Using (19), it is easily seen that (24) is quadratic in x. Defining

$$\alpha = g\,(V_G - V_B) - K/(1+\beta)$$

the two solutions are

$$x_2 = V_B + \frac{1}{2}\left[\alpha + \sqrt{\alpha^2 - 4(1-g)\,(V_G - V_B)\,K/(1+\beta)}\right] \text{ and } \quad x_1 \;= 2V_B + \alpha - x_2.$$

The corresponding market equilibria are obtained by reversing the transformation, which gives

$$\hat{E}_i = (x_i - p)/(1-\beta).$$

Using (19) and (24), the equilibrium level of honesty can be written

$$\hat{h}_{\hat{E}_i}(p) = \hat{h}_i = \frac{x_i - V_B + K/(1+\beta)}{g\,(V_G - V_B)}.$$

Proposition 7. A solution, x_i, to (24) is a market equilibrium if and only if $p \leq x_i$. There may be zero, one, or two market equilibria. If there is only one market equilibrium, it is generated by x_2. Given that $p \geq \theta$, there is one market equilibrium if $\hat{S}(p) \geq K$, and none otherwise. Given that $p < \theta$, there is one market equilibrium if $\hat{S}(p) > K$, two market equilibria if $\hat{S}(p) \leq K < \hat{S}(\theta)$, one if $\hat{S}(\theta) = K$, and none if $K > \hat{S}(\theta)$.

Proof. The structure of $\hat{S}$ was established in the proof of Proposition 4. In the transformed problem, the condition $0 \leq E \leq (V_H - p)/(1-\beta)$ becomes $p \leq x \leq V_H$, and the fact that $K \geq 0$, together with $\hat{S}$'s being nonnegative for feasible prices, guarantees the latter inequality. Since $\hat{S}$ is maximized at $\theta = p$, there can be no real solutions to (24) unless $K \leq \hat{S}(\theta)$. The conditions for zero, one, and two equilibria follow directly from the geometry of the situation. (See Figure 3.) Q.E.D.

Corollary 2. In the repeated game, sellers are honest if and only if the (net) buyers' search cost is zero.

Proof. Using (21), $\hat{h}(x) = 1$ if and only if $x = V_H$, and inspection of (25) reveals that $x = x_2 = V_H$ if and only if $K = 0$. Q.E.D.

Two intriguing features of the repeated game are that $\hat{h}_i$ is independent of the price and that there may be two distinct equilibria. The constancy of $\hat{h}_i$ comes about because the solution requires that $p + (1 - \beta)\hat{E}_i$ be a constant. Hence, higher prices lead directly to lower values of $\hat{E}_i$, leaving the opportunity cost, x_i, unchanged, which in turn leaves $\hat{h}_i$ unaltered. When there are two feasible solutions to the repeated game, there are two distinct levels of honesty (together with their associated terminal values) that make the buyers indifferent between repurchase and Switch. Producers are indifferent between $\hat{h}_1$ and $\hat{h}_2$. Buyers will, of course, prefer $\hat{h}_2$ to $\hat{h}_1$, but will play the same strategy in response to either.

Comparative statics analysis of two or more equilibria requires a convention for mapping equilibria onto the new equilibria that are generated by a parameter change. In what follows, we use the continuity of $\hat{E}_i$ and $\hat{h}_i$ to support the presumption that changes in a parameter can be mapped to changes in x_1 and x_2 and, hence, to changes in Equilibria 1 and 2. Observation 5. Increases in K produce decreases in $x_2, \hat{E}_2, \hat{h}_2$, and $\hat{T}_2$. Equilibrium 1, when it also exists, responds in the opposite manner: $x_1, \hat{E}_1, \hat{h}_1$, and $\hat{T}_1$ are increasing in K.

To gain intuition as to how (in Equilibrium 1) an increase in search costs can lead to a decrease in honesty, note that, ceterus paribus, an increase in K makes Rebuy more attractive. Therefore, to restore buyer indifference, h must be adjusted. Increases in h increase the values of both Rebuy and Switch, but in Equilibrium 1 the impact on Rebuy is more pronounced. Hence, to restore indifference, h must be reduced.

Competitive equilibrium. Competition in the repeated game takes place in each period. Buyers who have finished with a previous supplier, either after buying for one or both periods, seek the best price among those offered. In this setting, it is common knowledge that there is a stationary equilibrium in which each buyer takes the terminating value, $\hat{E}$, as fixed when evaluating the offers. Consequently, the competitive price in each period is that which maximizes $\hat{S}_E(p)$. Because $\hat{E}_i$ is strictly decreasing in price, the characterization of the competitive equilibrium is straightforward: the price falls to P_0. This argument establishes the competitive equilibrium.

Proposition 8. For each feasible solution to (23), x_i, there exists a single competitive equilibrium for the repeated game of $\left(P_0, \hat{h}_i; 0\right)$.

Note that even if the corresponding two-period game has a competitive equilibrium price $\theta > P_0$, the interior maximum is eliminated in the repeated game, and the competitive price drops to P_0.

In the interesting case in which there are two competitive equilibria, the first exhibits lower levels of honesty, consumer surplus, and total surplus than does the second; producer profits are the same in both. These two equilibria represent two internally consistent but different worlds. (Had the model contained more complex cost or technology specifications, we strongly suspect that there would be more than two equilibria.) In each, the buyers' ability to discipline a particular seller is aided or limited by the level of honesty known to be available from other sellers.

If search costs are zero, the competitive equilibrium in the repeated game produces the classic Klein-Leffler result. Sellers are perfectly disciplined by a boycott because a boycott is essentially costless-buyers can frictionlessly move to other sellers. But, if there are costs associated with changing from one seller to another, sellers are less than fully honest.

Discount strategies in the repeated game. Examining (22) and (21) reveals that introducing a first-period discount, d, will have the same effect on $\hat{E}_i$ and $\hat{h}_i$ as a reduction in search costs from

K to $K - d$.

Observation 6. A market equilibrium cannot have a first-period discount that is larger than K.

Intuitively, a first-period discount is simply a refund of search costs. Net search costs cannot be made negative, or all buyers will select Switch, never paying the higher secondperiod price. More formally, with discounts, the transformed fixed-point condition, (24), becomes $\hat{S}(x) = K - d$, which has real solutions if and only if $d \leq K$.

If $\hat{\Pi}(P_0) > K$, then the seller can clearly afford to pay a discount of $d = K$. This discount acts to eliminate buyers' search costs and, by Corollary 2, generates an equilibrium with completely honest sellers. But, an individual competitor need not stop there. To attract buyers away from other firms, a seller could offer a discount of up to $\hat{\Pi}(P_0)$. But, if each competitor does this, even the recipients of good products will choose Switch. To see this, note that $h = 1$ in the proposed equilibrium, so the buyers expect a surplus of $V_H - P_0$ in each period plus the bonus for switching, and that buyers will prefer to get the bonus (Switch) sooner rather than later. But, sellers, anticipating no second-period sales, will always choose Low, and the proposed equilibrium collapses. This argument, together with the fact that $\hat{\Pi}(P_0) = P_0 - C_L$, establishes the following proposition.

Proposition 9. There is a competitive market equilibrium with discounting in the repeated game if and only if $K \geq P_0 - C_L$.

Proof. The proof follows from the arguments given in the preceding paragraph. (A formal proof is available from the authors.)

The intuition behind this result is straightforward. If the competitive rents are very high, competitors will dissipate those rents in first-period giveaways. But, in the repeated game, buyers will realize that no one will ever rebuy in such a setting but will always prefer to seek a new seller's first-period giveaway. Thus, the market fails.

Given $K \geq P_0 - C_L$, the constraint that $d \leq K$ might be thought to limit the effectiveness of first-period discounts in the repeated game. Nevertheless, as in the two-period game, twoprice policies generate complete producer honesty and improve the total surplus and the consumer surplus.

Proposition 10. Define η as the unique price that satisfies $\hat{\Pi}(\eta) = K$, and define $\lambda = \max(\eta, P_0)$. If $K \leq \hat{\Pi}(V_H)$, the two-price repeated game has a unique competitive market equilibrium, $(K, \lambda, 1; \hat{r}(\lambda))$. If $K > \hat{\Pi}(V_H)$, then no market equilibrium exists. Proof. If $\hat{\Pi}(P_0) = K$, the producer can clearly afford to pay a discount of $d = K$. Such a policy eliminates Equilibrium 1 . Because $\hat{E}_2 > \hat{E}_1$ and because $\hat{E}_2$ decreases in K, the discount $d = K$ produces a greater consumer surplus than does any lower magnitude of the discount. Hence, competition will bring about discounts equal to the search cost, thereby generating the equilibrium $(0, K, P_0; 1)$.

Given that $\hat{\Pi}(P_0) < K$, the producer cannot afford a discount of K if the price remains at P_0. Given that $\hat{\Pi}(V_H) > K$, the fact that $\hat{\Pi}$ is strictly increasing ensures the existence of a unique η satisfying $\hat{\Pi}(\eta) = K$. To verify that the price will rise to η, suppose that the converse were true. Given a proposed equilibrium price $p < \eta$, each buyer and seller can take the terminating values, E_i, as fixed. But, the effective search costs are $K - \hat{\Pi}(p) > 0$. Therefore, $h_2 < 1$, which gives each individual producer the incentive (analyzed in Section 5) to raise the price, thereby offering greater honesty, and to refund the incremental profits as a first-period discount. Hence, the proposed equilibrium collapses. If $p = \eta$, then $h_2 = 1$; an individual producer's price increase cannot increase his honesty and, therefore, only guarantees the loss of second-period sales. Hence, price will rise to η.

If $K > \hat{\Pi}(V_H)$, then $x_2 > p$ for any feasible price, and there can be no market equilibrium. Q.E.D.

Discussion and Conclusions

Our purpose has been to study the effect of introducing performance uncertainty into a model of an experience good market, where quality is enforced by a repeat-purchase mechanism. The model we have constructed and analyzed highlights self-enforced exchange equilibria among rational players. Much of the strength of this approach lies in its clarity; there are no acts of nature or "types." Instead, the strategic behavior comes from a mutual understanding by firms and consumers of the driving forces that lie behind their self-interests. Despite the apparent simplicity of the model, the results are quite complex.

Our basic result is that the sellers will be fully honest only if the price rises to the consumers' reservation price. At that price, a buyer who has received a bad product but believes that the seller is honest is indifferent between repurchasing and doing without and can, therefore, forgo purchasing the good and enforce the equilibrium. Monopolists will automatically set this price and be honest. Competition also yields honest sellers if competitors can set the monopoly price in the second period and can offer deep discounts to first-period purchasers. However, if discounts or prices are restricted, the total surplus, consumer surplus, and sellers' honesty are reduced.

One example of a situation in which this outcome may arise in practice is government procurement, such as defense contracting. If products are difficult to fully evaluate upon delivery and if the seller's inputs are not easily observable, the conditional promise of repeat business may be the most effective means of assuring quality. However, if each new contract is awarded to the lowest bidder and if the buyer acts to limit the contractors' profits to "competitive" levels, then the value of repeat business is reduced, and rational sellers will devote less attention to quality. Note also that two-price schemes are normally not usable in this setting, as the buyer cannot agree to pay monopoly prices in the second period. Thus, legal and administrative restrictions, designed to prevent "profiteering," may actually reduce both the total and the consumer surplus.

When we allow for consumer switching by embedding the two-period game in a web of identical games and imposing a stationarity condition, a new pattern of equilibria emerges. In the repeated game, sellers are honest only if the buyer's net search cost is zero. If firstperiod discounts are feasible, competitors can generate the zero-search-cost condition by offering first-period discounts that equal the search costs. Interestingly, if competitors' rents are "too high," then their individual attempts to attract customers with discounts destroy the equilibrium; that is, if discounts exceed the search costs, buyers will always prefer to switch rather than rebuy, thus upsetting the repeat-purchase mechanism.

If there are level prices or restraints on discounts, the repeated game yields two equilibria, each having the same price and buyer strategy, but exhibiting different levels of seller honesty. In the "better" equilibrium, one might argue that sellers are kept fairly honest by the threat of a boycott; in the "poorer" equilibrium, one might note instead that a buyer's propensity to boycott is limited by the fact that other suppliers are equally dishonest. In this setting, competition can force a particular supplier to deliver normal levels of quality, but competition may not influence or explain the overall level of honesty or diligence.

References

and Faulhaber, G.R. "Optimism Invites Deception." Quarterly Journal of Economics, Vol. 103 (1988), pp. 397-407.

Grossman, S.J. and Perry, M. "Perfect Sequential Equilibrium." Journal of Economic Theory, Vol. 39 (1986), pp. 97-119.

Klein, B. and Leffler, K. "The Role of Market Forces in Assuring Contractual Performance." Journal of Political Economy, Vol. 89 (1981), pp. 615-641.
Kohlberg, E. and Mertens, J. "On the Strategic Stability of Equilibria." Econometrica, Vol. 54 (1986), pp. 1003-1037.
Kreps, D.M. and Wilson, R. "Sequential Equilibria." Econometrica, Vol. 50 (1982a), pp. 863-894.
and -. "Reputation and Imperfect Information." Journal of Economic Theory, Vol. 27 (1982b), pp. 253-279.
Nelson, P. "Information and Consumer Behavior." Journal of Political Economy, Vol. 78 (1970), pp. 311-329.
Rogerson, W. "Reputation and Product Quality." Bell Journal of Economics. Vol. 14 (1983), pp. 508-516.
Shapiro, C. "Consumer Information, Product Quality, and Seller Reputation." Bell Journal of Economics, Vol. 13 (1982), pp. 20-35.
——. "Optimal Pricing of Experience Goods." Bell Journal of Economics, Vol. 14 (1983a), pp. 497-507.
-. "Premiums for High Quality Products as Returns to
Reputations." Quarterly Journal of Economics, Vol. 98 (1983b), pp. 659-679. Stiglitz, J.E. "The Causes and Consequences of the Dependence of Quality on Price." Journal of Economic Literature, Vol. 25 (1987), pp. 1-48.
Telser, L. "A Theory of Self-Enforcing Agreements." Journal of Business, Vol. 53 (1980), pp. 27-44.

FIGURE 2: HONESTY AND TRUST AS FUNCTIONS OF PRICE

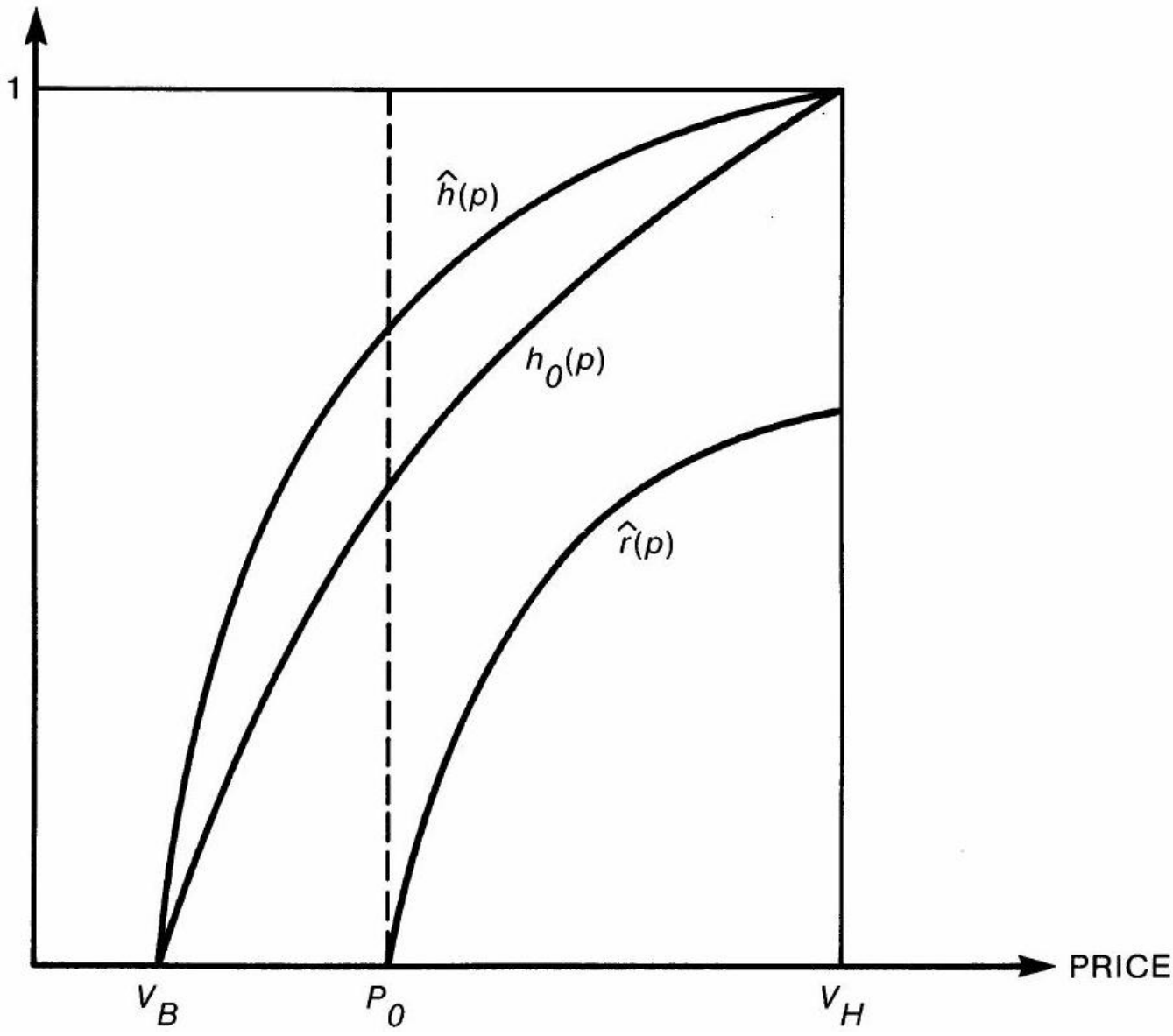
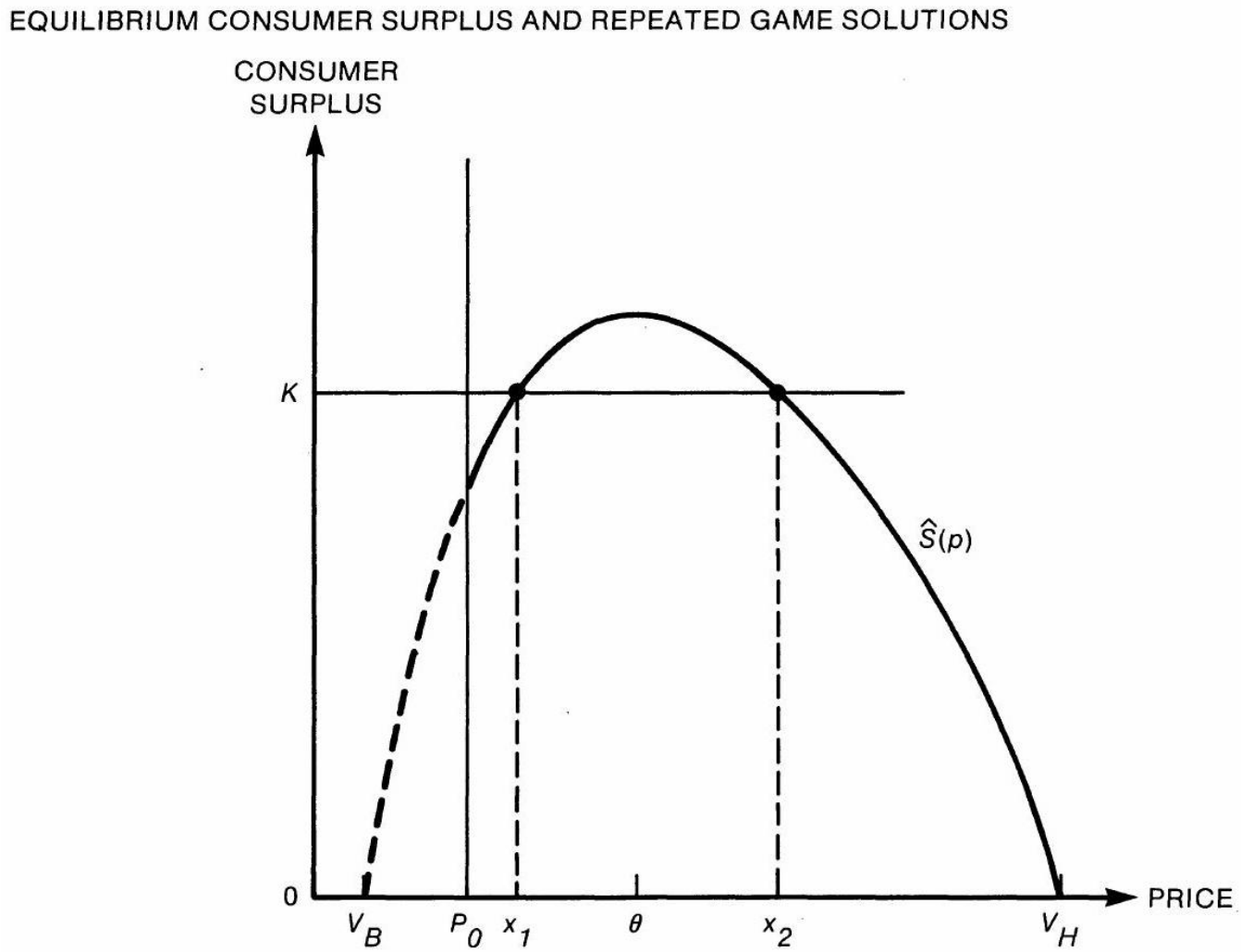


FIGURE 3:


EQUILIBRIUM CONSUMER SURPLUS AND REPEATED GAME SOLUTIONS
CONSUMER SURPLUS
K
0
$\hat{S}(p)$
PRICE
V_B
P_0
x_1
θ
x_2
V_H